

# Mathematics I

Complete beginner-friendly digital textbook for Course 1002. It works as textbook, revision guide, workbook, formula book and exam preparation guide in one offline HTML file.

Designed for first-semester diploma students who may be weak in mathematics. Learn the idea, copy the method, practise the example, then check the answer key.

## Course Details

**1002**

Course Code

**5**

Credits

**75**

Total Periods

**5-0-0**

L:T:P

**Basic Science**

Category

**Theory**

Subject Type

**Main areas:** Complex Numbers, Coordinate Geometry, Trigonometry, Limits, Differentiation I and Differentiation II.

## Course Overview

## What this subject teaches

### Why it matters

Complex numbers are used in electrical and electronics calculations. Straight lines help in graph interpretation. Trigonometry is used in waves, AC circuits, surveying and mechanics. Differentiation explains rate of change and slope.

### How to study

1. Memorise formula with meaning of symbols.
2. Write one solved example by hand.
3. Try the matching practice question.

## Objective

To provide introductory coverage of Trigonometry, Differential Calculus and basic elements of Algebra for diploma engineering students.

## Prerequisite

Secondary school algebra, factorisation, equations, coordinate plane, angle measurement, basic trigonometric ratios and simple graphs.

## Course outcomes and hours

| CO  | Outcome                                                               | Hours | Level    |
|-----|-----------------------------------------------------------------------|-------|----------|
| CO1 | Use complex numbers and equations of straight lines.                  | 20    | Applying |
| CO2 | Solve problems related to trigonometry.                               | 18    | Applying |
| CO3 | Use limits and derivatives to solve problems.                         | 20    | Applying |
| CO4 | Apply composite, parametric, implicit and successive differentiation. | 15    | Applying |

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Easy, medium, exam level

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Last-minute guide

### Answer Key

Final answers and hints

Module 1 • CO1 • 20 hours

## Complex Numbers and Straight Lines

Learn numbers of the form  $a+ib$ , then learn how straight lines are written, compared and solved in coordinate geometry.

## Complex numbers

### Definition, real part, imaginary part, $i$ and $i^2=-1$ close

A complex number is  $z=a+ib$ . Here  $a$  is real part and  $b$  is imaginary part. The symbol  $i$  is defined by  $i^2=-1$ .

**Formula:**  $\text{Re}(z)=a$ ,  $\text{Im}(z)=b$

**Basic:** For  $5-3i$ ,  $\text{Re}=5$  and  $\text{Im}=-3$ . **Exam:** If  $(2x+1)+i(y-4)=7+3i$ , then  $x=3$ ,  $y=7$ .

Mistake: imaginary part of  $5-3i$  is  $-3$ , not  $-3i$ .

Try: Find  $\text{Re}$  and  $\text{Im}$  of  $-2+9i$ .

## Equality, addition and subtraction close

Equal complex numbers have equal real parts and equal imaginary coefficients. Add or subtract real parts together and imaginary parts together.

$$(a+ib)+(c+id)=(a+c)+i(b+d)$$

$$(a+ib)-(c+id)=(a-c)+i(b-d)$$

**Basic:**  $(3+2i)+(5-7i)=8-5i$ . **Exam:**  $(8-3i)-(2+6i)=6-9i$ .

Mistake: Do not mix real part with imaginary coefficient.

Try:  $(4+9i)-(7-2i)$ .

## Multiplication, conjugate, modulus, amplitude, Cartesian and polar idea close

Multiply using brackets and replace  $i^2$  by  $-1$ . Conjugate changes sign of imaginary part. Modulus is distance from origin. Amplitude is angle idea. Polar conversion is excluded, so know the idea only.

$$(a+ib)(c+id)=(ac-bd)+i(ad+bc)$$

$$\bar{z}=a-ib, |z|=\sqrt{a^2+b^2}, \tan\theta=b/a$$

$$\text{Cartesian: } a+ib; \text{ Polar idea: } r(\cos\theta+i \sin\theta)$$

**Basic:**  $(2+3i)(4+i)=5+14i$ . **Exam:** For  $z=3+4i$ ,  $\bar{z}=3-4i$  and  $|z|=5$ .

Mistake:  $i^2$  is  $-1$ . Modulus is never negative.

Try: Find conjugate and modulus of  $8-15i$ .

## Straight lines

A straight line equation may be selected depending on the available data: slope, one point, two points, intercepts or general equation.

$$m = (y_2 - y_1) / (x_2 - x_1)$$

$$y = mx + c; y - y_1 = m(x - x_1)$$

$$(y - y_1) / (y_2 - y_1) = (x - x_1) / (x_2 - x_1)$$

$$x/a + y/b = 1; ax + by + c = 0$$

**Basic:** slope through (1,2),(3,6) is 2. **Exam:** slope 3 through (2,5):  $y - 5 = 3(x - 2)$ , so  $y = 3x - 1$ .

Mistake: Reversing only one difference changes sign.

Try: line with slope -2 through (1,4).

## Intersection, angle, parallel, perpendicular and distance close

Intersection satisfies both equations. Angle between lines uses their slopes. Parallel lines have equal slopes. Perpendicular lines have product -1. Distance formula gives shortest distance from point to line.

$$\tan \theta = |(m_2 - m_1) / (1 + m_1 m_2)|$$

**Parallel:**  $m_1 = m_2$ ; **Perpendicular:**  $m_1 m_2 = -1$

$$\text{Distance} = |ax_1 + by_1 + c| / \sqrt{a^2 + b^2}$$

**Basic:**  $x + y = 5$  and  $x - y = 1$  gives intersection (3,2). **Exam:** Distance from (2,-1) to  $5x - 12y + 6 = 0$  is 28/13.

Mistake: Use modulus in distance formula.

Try: Are slopes 3 and -1/3 perpendicular?

## Trigonometry

Trigonometry connects angles and ratios. Focus on formulas, standard values, signs and simple applications.

$$180^\circ = \pi \text{ rad}$$

Degree to radian:  $\theta\pi/180$

Radian to degree:  $\theta \times 180/\pi$

Calculator mode alert: degrees for degree problems, radians for limits.

## Ratios

$\sin\theta = \text{opposite/hypotenuse}$

$\cos\theta = \text{adjacent/hypotenuse}$

$\tan\theta = \text{opposite/adjacent}$

$\text{cosec} = 1/\sin$ ,  $\sec = 1/\cos$ ,  $\cot = 1/\tan$

## Standard values

| $\theta$     | $0^\circ$ | $30^\circ$   | $45^\circ$   | $60^\circ$   | $90^\circ$  |
|--------------|-----------|--------------|--------------|--------------|-------------|
| $\sin\theta$ | 0         | $1/2$        | $1/\sqrt{2}$ | $\sqrt{3}/2$ | 1           |
| $\cos\theta$ | 1         | $\sqrt{3}/2$ | $1/\sqrt{2}$ | $1/2$        | 0           |
| $\tan\theta$ | 0         | $1/\sqrt{3}$ | 1            | $\sqrt{3}$   | Not defined |

## Identities, allied and formula groups close

| Group           | Formula                                                                                                                                                                       | Use               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Identities      | $\sin^2 A + \cos^2 A = 1$ ; $1 + \tan^2 A = \sec^2 A$ ; $1 + \cot^2 A = \text{cosec}^2 A$                                                                                     | Simplification    |
| Allied          | $\sin(90^\circ - A) = \cos A$ ; $\cos(90^\circ - A) = \sin A$                                                                                                                 | Complement angles |
| Compound        | $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ ; $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ ; $\tan(A \pm B) = (\tan A \pm \tan B) / (1 \mp \tan A \tan B)$         | Exact values      |
| 2A and 3A       | $\sin 2A = 2 \sin A \cos A$ ; $\cos 2A = \cos^2 A - \sin^2 A$ ; $\tan 2A = 2 \tan A / (1 - \tan^2 A)$ ; $\sin 3A = 3 \sin A - 4 \sin^3 A$ ; $\cos 3A = 4 \cos^3 A - 3 \cos A$ | Multiple angles   |
| Product and sum | $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ ; $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$                                                                                         | Transformation    |

Sign memory: ASTC. I quadrant all positive, II sine, III tan, IV cos.

# Limits and Differentiation I

A function is an input-output rule. A limit is the value approached. A derivative is rate of change or slope.

## Function and limits

If  $f(x)=x^2+1$ , then  $f(3)=10$ . For limits, first try direct substitution. If  $0/0$  comes, simplify.

$$\lim_{x \rightarrow a} f(x)$$

$$\lim_{x \rightarrow 0} \sin x/x=1$$

$$\lim_{x \rightarrow 0} \tan x/x=1$$

## Differentiation by definition

$$f'(x)=\lim_{h \rightarrow 0} [f(x+h)-f(x)]/h$$

After understanding the definition, use standard derivative rules for simple problems.

## Standard derivatives

| y             | dy/dx             | Care          |
|---------------|-------------------|---------------|
| $x^n$         | $nx^{n-1}$        | Power rule    |
| $\sin x$      | $\cos x$          | Radian idea   |
| $\cos x$      | $-\sin x$         | Negative sign |
| $\tan x$      | $\sec^2 x$        | Standard      |
| $e^x$         | $e^x$             | Same          |
| $\log x$      | $1/x$             | Natural log   |
| $\sin^{-1} x$ | $1/\sqrt{1-x^2}$  | Inverse trig  |
| $\cos^{-1} x$ | $-1/\sqrt{1-x^2}$ | Negative      |
| $\tan^{-1} x$ | $1/(1+x^2)$       | Inverse trig  |

$$d(u \pm v)/dx = du/dx \pm dv/dx$$

$$d(ku)/dx = k du/dx$$

$$d(uv)/dx = u dv/dx + v du/dx$$

$$d(u/v)/dx = (v du/dx - u dv/dx)/v^2$$

Common errors: wrong substitution in limits, missing brackets, derivative of  $\cos x$  without minus sign, wrong product rule.

Module 4 • CO4 • 15 hours

## Differentiation II

Use chain rule for function of a function, parametric derivative for  $x$  and  $y$  in  $t$ , implicit differentiation for mixed  $x$ - $y$  equations and successive differentiation up to second order.

### Chain rule

$$\text{If } y=f(u), u=g(x), dy/dx = dy/du \times du/dx$$

Flow: outer derivative  $\times$  inner derivative. Avoid highly nested functions excluded in syllabus.

### Parametric

$$dy/dx = (dy/dt)/(dx/dt)$$

Differentiate  $y$  and  $x$  separately with respect to  $t$ , then divide.

### Implicit

$$d(y^2)/dx = 2y dy/dx$$

When  $x$  and  $y$  are mixed, differentiate both sides and collect  $dy/dx$  terms.



$dy/dx$  = first derivative;  $d^2y/dx^2$  = second derivative

### Successive differentiation table

| y              | dy/dx           | d <sup>2</sup> y/dx <sup>2</sup> |
|----------------|-----------------|----------------------------------|
| x <sup>4</sup> | 4x <sup>3</sup> | 12x <sup>2</sup>                 |
| sinx           | cosx            | -sinx                            |
| cosx           | -sinx           | -cosx                            |
| e <sup>x</sup> | e <sup>x</sup>  | e <sup>x</sup>                   |

#### Formula Bank

## All formulas in one place

| Area       | Formula                               | Where used          | Mini example            | Mistake                 |
|------------|---------------------------------------|---------------------|-------------------------|-------------------------|
| Complex    | $z=a+ib$                              | Identify parts      | $3+4i$                  | Im is 4, not 4i         |
| Conjugate  | $\bar{z}=a-ib$                        | Change sign         | $2-5i \Rightarrow 2+5i$ | Changing real sign      |
| Modulus    | $ z =\sqrt{(a^2+b^2)}$                | Magnitude           | $ 3+4i =5$              | Negative answer         |
| Slope      | $m=(y_2-y_1)/(x_2-x_1)$               | Line steepness      | $m=2$                   | Wrong order             |
| Line       | $y=mx+c$ ; $y-y_1=m(x-x_1)$           | Equation of line    | $y=2x+1$                | Wrong c                 |
| Distance   | $ ax_1+by_1+c /\sqrt{(a^2+b^2)}$      | Point to line       | $3x+4y-5$               | No modulus              |
| Conversion | $180^\circ=\pi$ rad                   | Angle units         | $60^\circ=\pi/3$        | Mode error              |
| Identity   | $\sin^2A+\cos^2A=1$                   | Simplify            | $1-\cos^2A=\sin^2A$     | No square               |
| Compound   | $\sin(A+B)=\sin A\cos B+\cos A\sin B$ | Exact value         | $\sin 75^\circ$         | Sign                    |
| Double     | $\sin 2A=2\sin A\cos A$               | 2A problems         | $A=30^\circ$            | Wrong formula           |
| Limit      | $\lim \sin x/x=1$                     | Trig limits         | $x \rightarrow 0$       | Use radians             |
| Power      | $d(x^n)/dx=nx^{n-1}$                  | Derivative          | $x^5 \Rightarrow 5x^4$  | $nx^n$                  |
| Product    | $d(uv)=u \, dv+v \, du$               | Product             | $x\sin x$               | Multiplying derivatives |
| Quotient   | $d(u/v)=(vdu-udv)/v^2$                | Fraction            | $x/(x+1)$               | Wrong order             |
| Chain      | $dy/dx=dy/du \times du/dx$            | Composite           | $(3x+2)^4$              | Missing inner           |
| Parametric | $dy/dx=(dy/dt)/(dx/dt)$               | x,y in t            | $x=t^2,y=t^3$           | Reverse division        |
| Second     | $d^2y/dx^2$                           | Differentiate twice | $x^3 \Rightarrow 6x$    | Stop early              |

#### Solved Examples Bank

## 38 solved examples

**M1-E1 Add**

$3+4i$  and  $5-7i$ .

Real  $3+5=8$ , imaginary  $4-7=-3$ .

$$8-3i$$

**M1-E2 Subtract**

$(6-2i)-(1+9i)$ .

$6-1=5$ ,  $-2-9=-11$ .

$$5-11i$$

**M1-E3 Multiply**

$(2+3i)(4-i)$ .

$8-2i+12i-3i^2=11+10i$ .

$$11+10i$$

**M1-E4 Conjugate**

$z=7-24i$ .

$\bar{z}=7+24i$ ,  $|z|=\sqrt{625}$ .

$$\bar{z}=7+24i, |z|=25$$

**M1-E5 Equality**

$x+2i=5+yi$ .

Equate real and imaginary parts.

$$x=5, y=2$$

**M1-E6 Slope**

$(2,3), (6,11)$ .

$m=(11-3)/(6-2)$ .

$$m=2$$

**M1-E7 Line**

Slope 4 through  $(1,2)$ .

$y-2=4(x-1)$ .

$$y=4x-2$$

**M1-E8 Two-point line**

$(1,1), (3,5)$ .

$m=2$ ,  $y-1=2(x-1)$ .

$$y=2x-1$$

**M1-E9 Intersection**

$x+y=7$ ,  $x-y=1$ .

Add equations:  $2x=8$ .

$$(4,3)$$

**M1-E10 Distance**

$(2,1)$  to  $3x+4y-10=0$ .

$|6+4-10|/5$ .

$$0$$

**M2-E1 Degree to radian**

$150^\circ$ .

$150\pi/180$ .

$$5\pi/6$$

**M2-E2 Radian to degree**

$2\pi/3$ .

$(2\pi/3) \times 180/\pi$ .

$$120^\circ$$

**M2-E3 Standard value**

$\sin 60^\circ + \cos 60^\circ$ .

$\sqrt{3}/2 + 1/2$ .

$$(\sqrt{3}+1)/2$$

**M2-E4 Identity**

$1-\cos^2 A$ .

Use  $\sin^2 A + \cos^2 A = 1$ .

$$\sin^2 A$$

**M2-E5 Allied**

$\sin(90^\circ - A)$ .

Complement formula.

$$\cos A$$

**M2-E6 Compound** $\sin 75^\circ$ .

$$\sin(45+30)=\sin 45 \cos 30 + \cos 45 \sin 30.$$

$$(\sqrt{6}+\sqrt{2})/4$$

**M2-E7 Difference** $\cos 15^\circ$ .

$$\cos(45-30)=\cos 45 \cos 30 + \sin 45 \sin 30.$$

$$(\sqrt{6}+\sqrt{2})/4$$

**M2-E8 Double angle** $\sin A=3/5$ ,  $\cos A=4/5$ .

$$\sin 2A=2\sin A \cos A.$$

$$24/25$$

**M2-E9 Triple angle** $\sin 3A$ .

Use standard formula.

$$3\sin A - 4\sin^3 A$$

**M2-E10 Product** $2\sin A \cos B$ .

Product-to-sum formula.

$$\sin(A+B) + \sin(A-B)$$

**M3-E1 Function** $f(x)=x^2+2x$ ,  $f(3)$ .

$$9+6.$$

$$15$$

**M3-E2 Direct limit** $\lim_{x \rightarrow 2} (x^2+3)$ .

Substitute 2.

$$7$$

**M3-E3 Algebraic limit** $\lim_{x \rightarrow 1} (x^2-1)/(x-1)$ .Factor to  $x+1$ .

$$2$$

**M3-E4 Trig limit** $\lim_{x \rightarrow 0} \sin x/x$ .

Standard limit.

$$1$$

**M3-E5 Power** $d(x^5)/dx$ .

Power rule.

$$5x^4$$

**M3-E6 Polynomial** $d(x^3+4x^2-7)/dx$ .

Differentiate termwise.

$$3x^2+8x$$

**M3-E7 Trig derivative** $d(\sin x + \cos x)/dx$ .

$$\cos x - \sin x.$$

$$\cos x - \sin x$$

**M3-E8 Product** $d(x \sin x)/dx$ .

$$x \cos x + \sin x.$$

$$x \cos x + \sin x$$

**M3-E9 Quotient** $d[x/(x+1)]/dx$ .

$$[(x+1)-x]/(x+1)^2.$$

$$1/(x+1)^2$$

**M3-E10 Exp-log** $d(e^x + \log x)/dx$ .

Use standard derivatives.

$$e^x + 1/x$$

**M4-E1 Chain**

$$d(3x+2)^4/dx.$$

$$4(3x+2)^3 \times 3.$$

$$12(3x+2)^3$$

**M4-E2 Chain trig**

$$d \sin(5x)/dx.$$

$$\cos(5x) \times 5.$$

$$5\cos(5x)$$

**M4-E3 Parametric**

$$x=t^2, y=t^3.$$

$$dy/dx=(3t^2)/(2t).$$

$$3t/2$$

**M4-E4 Parametric**

$$x=2t+1, y=t^2.$$

$$dy/dx=2t/2.$$

$$t$$

**M4-E5 Implicit**

$$x^2+y^2=25.$$

$$2x+2y \, dy/dx=0.$$

$$dy/dx=-x/y$$

**M4-E6 Implicit**

$$xy=10.$$

$$x \, dy/dx+y=0.$$

$$dy/dx=-y/x$$

**M4-E7 Second derivative**

$$y=x^4.$$

$$y'=4x^3, \, y''=12x^2.$$

$$12x^2$$

**M4-E8 Second derivative**

$$y=\sin x.$$

$$y'=\cos x, \, y''=-\sin x.$$

$$-\sin x$$

**Practice Questions****Workbook practice****Easy****Module 1**

- Find Re and Im of  $8-11i$ .
- Add  $7+2i$  and  $3-9i$ .
- Find conjugate of  $-4+6i$ .
- Find slope through  $(0,2)$  and  $(4,10)$ .

**Medium****Module 1**

- Multiply  $(3+2i)(5-i)$ .
- Find line through  $(2,3)$  with slope  $-1$ .
- Find intersection of  $x+y=9$  and  $x-y=3$ .
- Find distance from  $(1,1)$  to  $3x+4y-12=0$ .

**Exam Level****Module 2**

- Convert  $225^\circ$  into radians.
- Find exact value of  $\cos 75^\circ$ .
- Simplify  $(1-\sin^2 A)/\cos A$ .
- Transform  $2\cos A \cos B$ .

**Medium****Module 3**

- Evaluate  $\lim_{x \rightarrow 2} (x^2-4)/(x-2)$ .

**Exam Level****Module 4**

- Differentiate  $(2x-1)^5$ .

**Mixed****Quick Check**

- State product rule.

2. Differentiate  $x^4 - 5x^2 + 6$ .

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3. Differentiate  $x \cos x$ .

4. Differentiate  $(x^2 + 1)/x$ .

2. If  $x = t^2 + 1$  and  $y = t^3$ , find  $dy/dx$ .

3. Differentiate  $x^2 + xy + y^2 = 7$  implicitly.

4. Find second derivative of  $y = \cos x + x^3$ .

2. State quotient rule.

3. Write  $\sin 2A$  formula.

4. Condition for perpendicular lines?

### Quick Revision

## Last-minute revision

### Must know

- $i^2 = -1$ ,  $|a + ib| = \sqrt{a^2 + b^2}$ .
- Slope, line forms, angle and distance formulas.
- $180^\circ = \pi$  rad and standard angle table.
- Direct substitution, factorisation and standard trig limits.
- Power, product, quotient and chain rules.

### Common mistakes

- Wrong sign in  $\cos$  derivative.
- Using degree mode for calculus limits.
- Forgetting inner derivative in chain rule.
- Not using  $dy/dx$  for  $y$  terms in implicit differentiation.
- Stopping at first derivative when second derivative is asked.

### Master Answer Key

## Answers for practice questions

**M1 Easy:** 1)  $\text{Re}=8$   $\text{Im}=-11$ . 2)  $10-7i$ . 3)  $-4-6i$ . 4)  $m=2$ .

**M1 Medium:** 1)  $17+7i$ . 2)  $y=-x+5$ . 3)  $(6,3)$ . 4)  $\text{distance}=1$ .

**M2:** 1)  $5\pi/4$ . 2)  $(\sqrt{6}-\sqrt{2})/4$ . 3)  $\cos A$ . 4)  $\cos(A+B) + \cos(A-B)$ .

**M3:** 1) 4. 2)  $4x^3 - 10x$ . 3)  $\cos x - x \sin x$ . 4)  $1 - 1/x^2$ .

**M4:** 1)  $10(2x-1)^4$ . 2)  $3t/2$ . 3)  $dy/dx = -(2x+y)/(x+2y)$ . 4)  $-\cos x + 6x$ .

**Mixed:** Product rule  $d(uv)=u dv+v du$ . Quotient rule  $d(u/v)=(vdu-udv)/v^2$ .  $\sin 2A=2\sin A\cos A$ . Perpendicular condition  $m_1m_2=-1$ .